

Chained SC Decoding for MAC Polar Codes at Corner Rate Points

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1. System Model

Two users transmit binary codewords $\mathbf{x}, \mathbf{y} \in \{0, 1\}^N$ over a multiple access channel (MAC). The channel output is

$$Z_i = f(X_i, Y_i, W_i), \quad i = 1, \dots, N$$

where W_i is the channel noise. For the Gaussian MAC (GMAC):

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + W_i, \quad W_i \sim \mathcal{N}(0, \sigma^2)$$

Each user encodes independently with a polar code: $\mathbf{x} = \mathbf{u}B_N F^{\otimes n}$ and $\mathbf{y} = \mathbf{v}B_N F^{\otimes n}$, where $\mathbf{u}, \mathbf{v} \in \{0, 1\}^N$ are the message vectors, $F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, and B_N is the bit-reversal permutation.

2. Monotone Chain Paths and Corner Rates

The MAC capacity region is achieved by decoding the $2N$ message bits $(u_1, \dots, u_N, v_1, \dots, v_N)$ in a specific order, defined by a **path** $\mathbf{b} \in \{0, 1\}^{2N}$, where $b_t = 0$ means “decode a U -bit” and $b_t = 1$ means “decode a V -bit.”

A monotone chain path has the form $\mathbf{b} = 0^i 1^N 0^{N-i}$ for some $0 \leq i \leq N$. The **corner rate points** are:

- **Class A** ($i = 0$): path $1^N 0^N$ — decode all of V first, then all of U
- **Class C** ($i = N$): path $0^N 1^N$ — decode all of U first, then all of V

3. Class C Decomposition

For Class C (path $0^N 1^N$), all N bits of U are decoded before any bit of V . This creates a natural two-stage decomposition:

Stage 1 — Decode U on the marginal channel:

When decoding U , no information about V is available. The decoder sees the MAC output $\mathbf{z}$ and treats Y as noise. The effective channel for X is:

$$P(Z|X) = \sum_{Y \in \{0,1\}} P(Z|X, Y) \cdot P(Y) = \frac{1}{2}P(Z|X, Y=0) + \frac{1}{2}P(Z|X, Y=1)$$

This is a **binary-input mixture channel**. For the GMAC:

$$Z|X \sim \frac{1}{2}\mathcal{N}(1 - 2X + 1, \sigma^2) + \frac{1}{2}\mathcal{N}(1 - 2X - 1, \sigma^2)$$

Stage 1 applies standard single-user SC decoding of $\mathbf{u}$ on this marginal channel. The frozen set $\mathcal{F}_U$ is designed for this channel.

Stage 2 — Decode V on the clean channel:

After Stage 1, we have $\hat{\mathbf{u}}$ and can reconstruct $\hat{\mathbf{x}} = \hat{\mathbf{u}}B_N F^{\otimes n}$. Now the decoder sees both $\mathbf{z}$ and $\hat{\mathbf{x}}$. The effective channel for Y is:

$$P(Z|Y, \hat{X}) = P(Z|\hat{X}, Y) \quad (\text{assuming } \hat{X} = X)$$

If Stage 1 decoded correctly ($\hat{\mathbf{x}} = \mathbf{x}$), this is a **clean single-user channel** — the interference from X is perfectly subtracted. For the GMAC:

$$Z - (1 - 2\hat{X}) = (1 - 2Y) + W \quad \Rightarrow \quad \tilde{Z}|Y \sim \mathcal{N}(1 - 2Y, \sigma^2)$$

This is just a standard BPSK-AWGN channel. Stage 2 applies single-user SC decoding of $\mathbf{v}$ on this channel.

4. Chained SC Decoder

The full decoder chains the two stages:

1. **Compute channel features:** From $\mathbf{z}$, compute per-position LLRs for the mixture channel:

$$\Lambda_i^{(1)} = \log \frac{P(Z_i|X_i = 0)}{P(Z_i|X_i = 1)} = \log \frac{P(Z_i|X=0, Y=0) + P(Z_i|X=0, Y=1)}{P(Z_i|X=1, Y=0) + P(Z_i|X=1, Y=1)}$$

2. **Stage 1 SC decode:** Feed $\Lambda^{(1)}$ to single-user SC decoder $\rightarrow \hat{\mathbf{u}}$

3. **Reconstruct:** $\hat{\mathbf{x}} = \hat{\mathbf{u}}B_N F^{\otimes n}$

4. **Compute clean LLRs:**

$$\Lambda_i^{(2)} = \log \frac{P(Z_i|\hat{X}_i, Y_i = 0)}{P(Z_i|\hat{X}_i, Y_i = 1)}$$

$$\text{For GMAC: } \Lambda_i^{(2)} = \frac{2(Z_i - (1 - 2\hat{X}_i))}{\sigma^2}$$

5. **Stage 2 SC decode:** Feed $\Lambda^{(2)}$ to single-user SC decoder $\rightarrow \hat{\mathbf{v}}$

5. Error Analysis

The total block error rate is bounded by:

$$P_e^{\text{total}} \leq P_e^{(1)} + P_e^{(2)} + P_e^{\text{cascade}}$$

where $P_e^{(1)}$ is the Stage 1 error rate on the mixture channel, $P_e^{(2)}$ is the Stage 2 error rate on the clean channel (conditioned on correct Stage 1), and P_e^{cascade} accounts for error propagation when $\hat{\mathbf{x}} \neq \mathbf{x}$.

In practice, $P_e^{(2)} \ll P_e^{(1)}$ because the clean channel (with interference removed) is much better than the mixture channel. The dominant term is $P_e^{(1)}$.

6. Neural Chained Decoder

The chained NPD replaces both SC decoders with Neural Polar Decoders (NPDs), trained independently:

- **Stage 1 NPD:** Trained on the mixture channel with fast parallel cross-entropy (fast_ce), $O(\log N)$ gradient depth
- **Stage 2 NPD:** Trained on the clean channel (given correct $\hat{\mathbf{x}}$), same fast_ce training

Each NPD uses the same architecture: z -encoder $\rightarrow$ recursive CheckNode/BitNode tree $\rightarrow$ leaf decisions. The key advantage is **NPD-guided code design**: train at rate 1, measure per-position mutual information, select information positions where the NPD performs best (which may differ from the SC-optimal positions).